

Case Study: Cross-Channel Marketing Analytics Dashboard

1. Introduction

Marketing teams today manage campaigns across multiple platforms, from Google Ads to Facebook Ads, alongside website analytics. However, reporting is often siloed and requires extensive manual effort. The goal of this project was to build an automated data pipeline and interactive dashboards that provide a unified, real-time view of campaign performance across channels.

2. Problem Statement

Marketing stakeholders faced three main challenges:

- Disparate datasets across ad platforms and web analytics, making it hard to consolidate performance.
- Manual reporting workflows consuming 10–15 hours per week.
- Limited visibility into KPIs such as CTR, CPC, CPA, ROAS, and funnel conversion rates.

3. Solution Approach

To address these challenges, I designed and implemented a cross-channel marketing analytics solution with the following steps:

- Built an ETL pipeline in Python to extract Google Ads, Facebook Ads, and web analytics data.
- Applied transformations and implemented QA checks to ensure accuracy and consistency.
- Automated daily refresh of reporting tables using GitHub Actions, outputting data to Google Sheets.
- Designed Tableau and Looker Studio dashboards with KPI tiles, funnels, cross-channel comparisons, and device-level analysis.

4. Tools & Technologies

- Python, Pandas – ETL and QA checks
- Google Ads API, Facebook Marketing API – Data extraction
- GitHub Actions – Workflow automation for daily refresh
- Google Sheets – Central data store for dashboards
- Tableau & Looker Studio – Dashboard design and data visualization

5. Results & Impact

- Reduced manual reporting workload by 80%, saving ~15 hours per week.
- Provided real-time visibility into campaign performance, enabling faster optimization decisions.
- Improved accuracy of reports by 15% through built-in QA checks.
- Delivered executive-ready dashboards tailored for both marketers and leadership teams.

6. Visuals & Dashboards

The following figures showcase examples of the dashboards developed as part of this project:

[Insert Screenshot: Tableau KPI Dashboard]

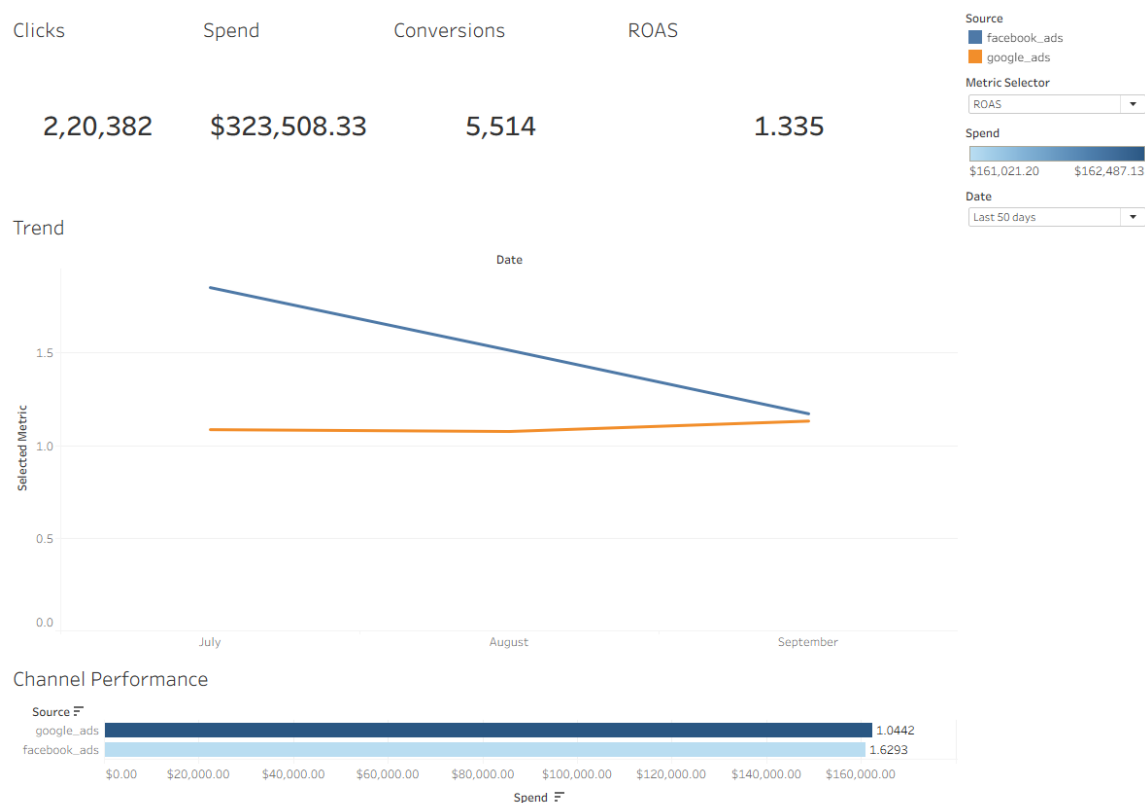


Figure 1: Tableau KPI Dashboard

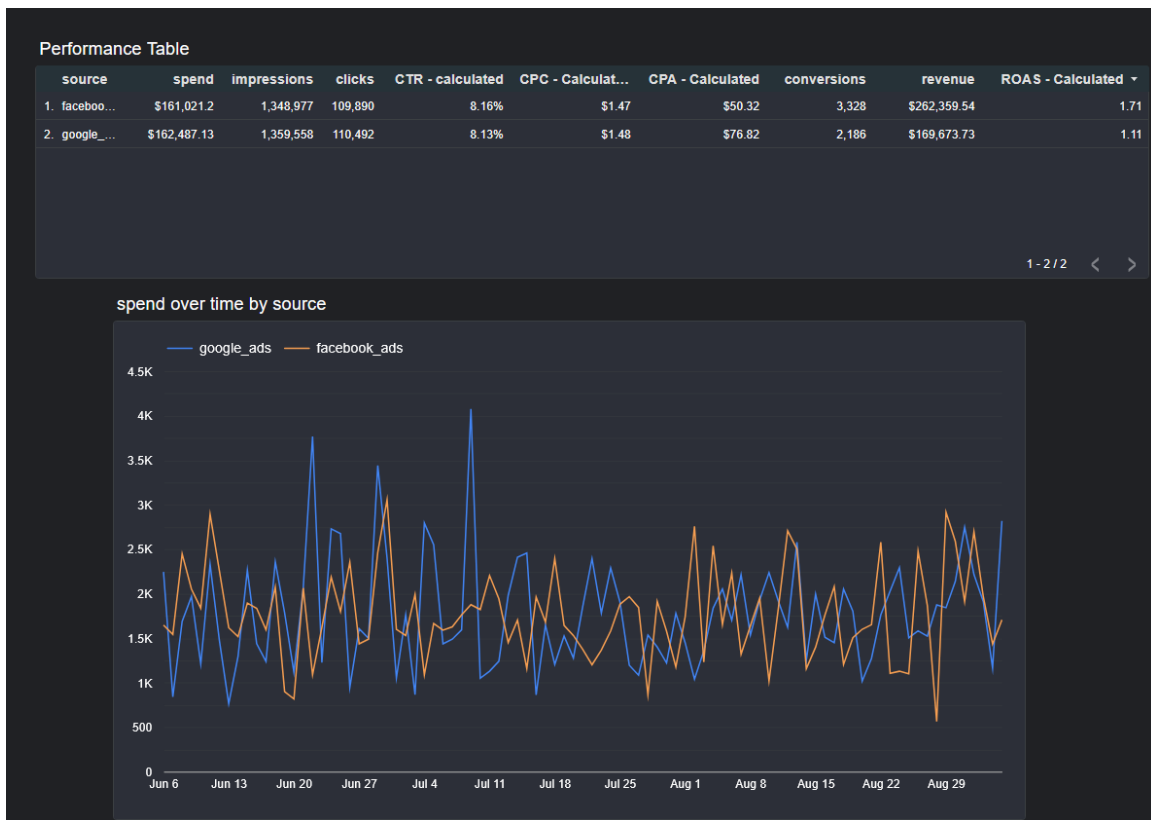


Figure 2 : Looker - Channel Performance Comparison

7. Conclusion

This project demonstrates the value of integrating ETL pipelines with BI tools to streamline marketing analytics. By unifying data sources, automating reporting, and visualizing KPIs effectively, the solution not only saves time but also improves decision-making speed and accuracy. Future improvements may include expanding data storage to BigQuery, integrating predictive models for ROI forecasting, and embedding AI-driven recommendations for budget optimization.